

Corrected Mode-I Adaptive Remeshing: Localization and Fracture-Response Verification

Focused Supervisor Report – 08 October 2026

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1. Objective and decision context

The purpose of this short report is to answer one focused question: *did the corrected native Abaqus remeshing workflow localize refinement to the Mode-I crack-propagation corridor while preserving an acceptable fracture response?* The work therefore has three objectives:

1. identify why the earlier adaptive mesh refined too much of the domain;
2. demonstrate the corrected, crack-path-localized mesh; and
3. verify the corrected mesh using reaction force, initial stiffness, spatial damage, and energy evidence.

Main outcome. The earlier Step-1/pre-peak remeshing state produced broad refinement, whereas evaluation of the localized Step-2 field produced a narrow horizontal corridor. The governed ET1 corrected mesh contains 14,483 finite elements, with 64.12% of them in $|y - 0.5| \leq 0.05$ mm. It preserves the initial stiffness to -0.0261% relative to the fixed reference and predicts $F_{\max}$ within -1.8563% .

Table 1: Supervisor-facing status at the start of the 08 October meeting.

Topic	Status	Evidence-based interpretation
Earlier adaptive mesh	Superseded for localization assessment	Native remeshing worked, but the selected Step-1 field drove broad refinement.
Corrected Step-2 mesh	Verified	Refinement follows the horizontal ligament and avoids refinement on the initial slit flanks.
Mechanical response	Verified over tested cases	K_0 , $F_{\max}$, u_{peak} , and crack symmetry remain stable or acceptably close to the fixed reference.
Energetic interpretation	Qualified with limitation	Component histories are available; the exact global energy identity remains not yet closed.
Gate status	Ready for supervisor sign-off	Gate 6C, Mode-II, and Gate 7 remain on hold until a formal decision.

This report deliberately omits the detailed gate ledger, long derivations, intermediate failed packages, and full job history. Those remain in the governed project records.

2. Problem with the earlier adaptive mesh

The earlier native adaptive mesh was operational, but it was not an efficient representation of the intended fracture process zone. The 1% target-error case contained 71,320 finite elements and refined a large part of the specimen rather than concentrating resolution primarily along the future horizontal crack path. Figure 1 retains the full-domain and crack-tip views; the first column is the relevant 71,320-element case.

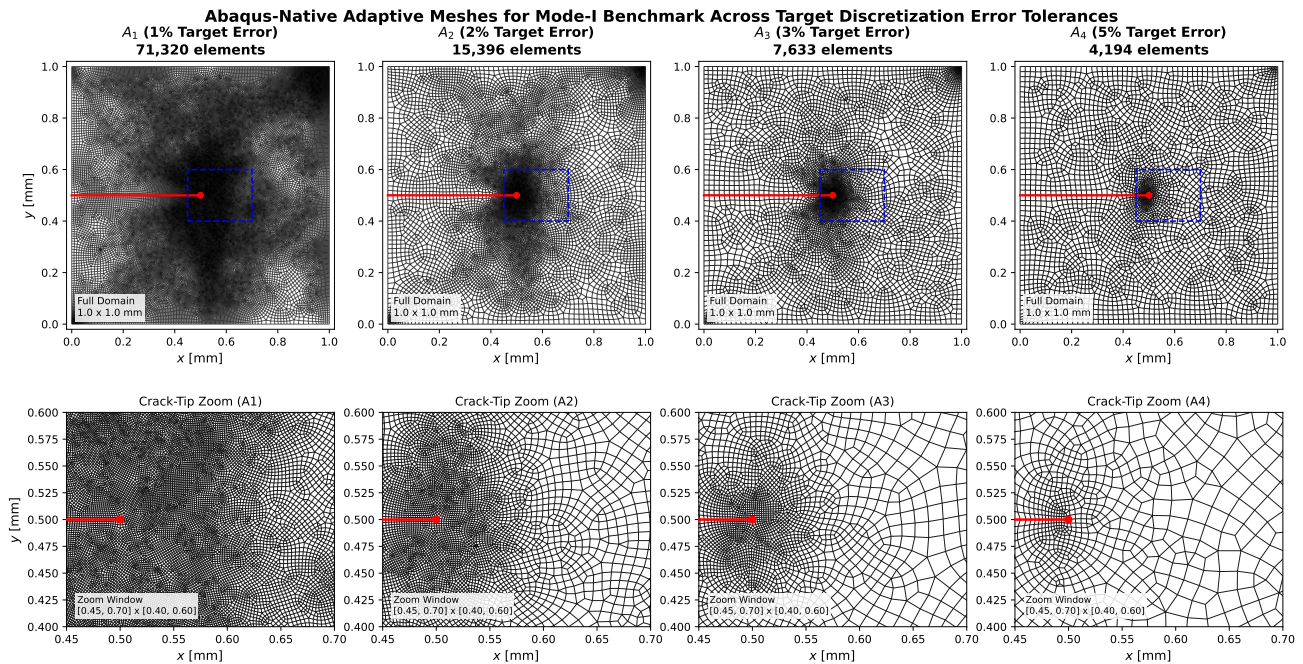


Figure 1: Earlier Step-1/pre-peak adaptive meshes. The A_1 case in the first column shows the 71,320-element full-domain mesh and its crack-tip zoom. Although the crack tip is refined, the refinement is also distributed broadly through the specimen. The other columns show that relaxing `errorTarget` reduces the element count but does not by itself correct the physical-stage selection.

Scientific conclusion. Element count alone was not a sufficient quality measure. The central problem was spatial selectivity: too much resolution was placed away from the propagating ligament.

The broad pattern had two distinct issues that must not be conflated. First, an early boundary-set formatting defect omitted bottom-edge constraints and distorted stiffness; that defect was corrected and closed. Second, even with correct boundary conditions, the remeshing field had been evaluated at a Step-1/pre-peak state, before the fracture process had localized along the ligament. The second issue explains the inefficient mesh morphology addressed here.

3. Corrections introduced

The decisive change was to align the remeshing evaluation state with the physical feature that the mesh is intended to resolve. The native **RemeshingRule** now evaluates the localized Step-2 companion field instead of treating an early pre-peak error field as the final sizing basis.

Table 2: Difference table: earlier state, implemented correction, and observed effect.

Item	Earlier state / issue	Correction applied	Observed effect
Remeshing evaluation state	Step-1/pre-peak field emphasized the initial crack-tip neighborhood and broad background error.	Evaluate the localized Step-2 field after the crack-propagation pattern is established.	Refinement extends along the horizontal ligament instead of filling most of the domain.
Boundary-condition set	A long N_BOTTOM card was parsed incompletely in an early model.	Wrap node identifiers to valid Abaqus keyword lines.	All intended supports are applied; K_0 returns to the reference range.
Quality criterion	Total element count received excessive emphasis.	Judge corridor share, band width, far-field fraction, crack symmetry, and response together.	Mesh efficiency is assessed by physical localization rather than count matching alone.
Tolerance study	A single tolerance could hide morphology sensitivity.	Evaluate errorTarget $\in \{1, 2, 3, 5\}\%$ with identical bounds and region.	ET1 is target-like; ET2–ET5 progressively lose ligament resolution.
Post-processing	Results were dispersed across separate checks.	Use single-job provenance for RF- u , K_0 , energy, phase field, and crack-path metrics.	Mechanical and spatial conclusions can be traced to exact jobs without cross-copying.

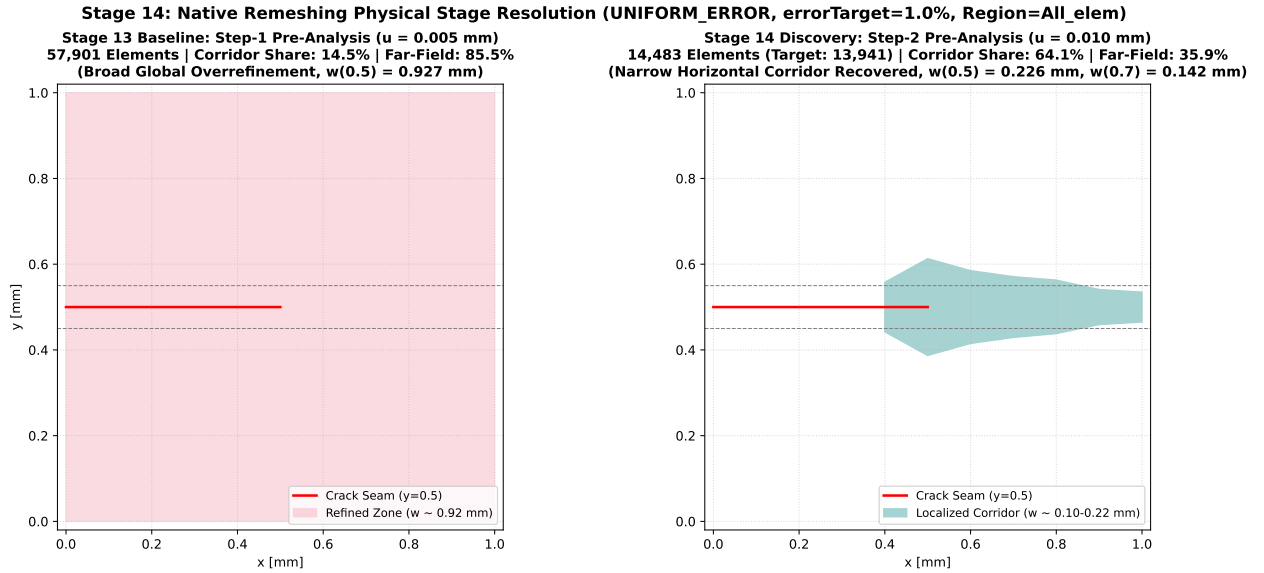


Figure 2: Isolated effect of physical-stage selection at **errorTarget=1%**. The Step-1/pre-peak evaluation produces broad global refinement, while the corrected Step-2 evaluation recovers a horizontal corridor. The corrected mesh has 14,483 elements, 64.1% corridor share, and 35.9% far-field share.

Scientific conclusion. The main localization improvement comes from evaluating the physically relevant Step-2 field, not from tuning the element count after the fact.

4. Corrected adaptive mesh

Figure 3 shows the governed corrected ET1 mesh at both specimen and ligament scales. It contains 14,483 finite elements (14,082 quadrilaterals and 401 triangles) and 14,456 finite-element nodes. The minimum and median area-equivalent element sizes are $0.760\ \mu\text{m}$ and $2.597\ \mu\text{m}$, respectively.

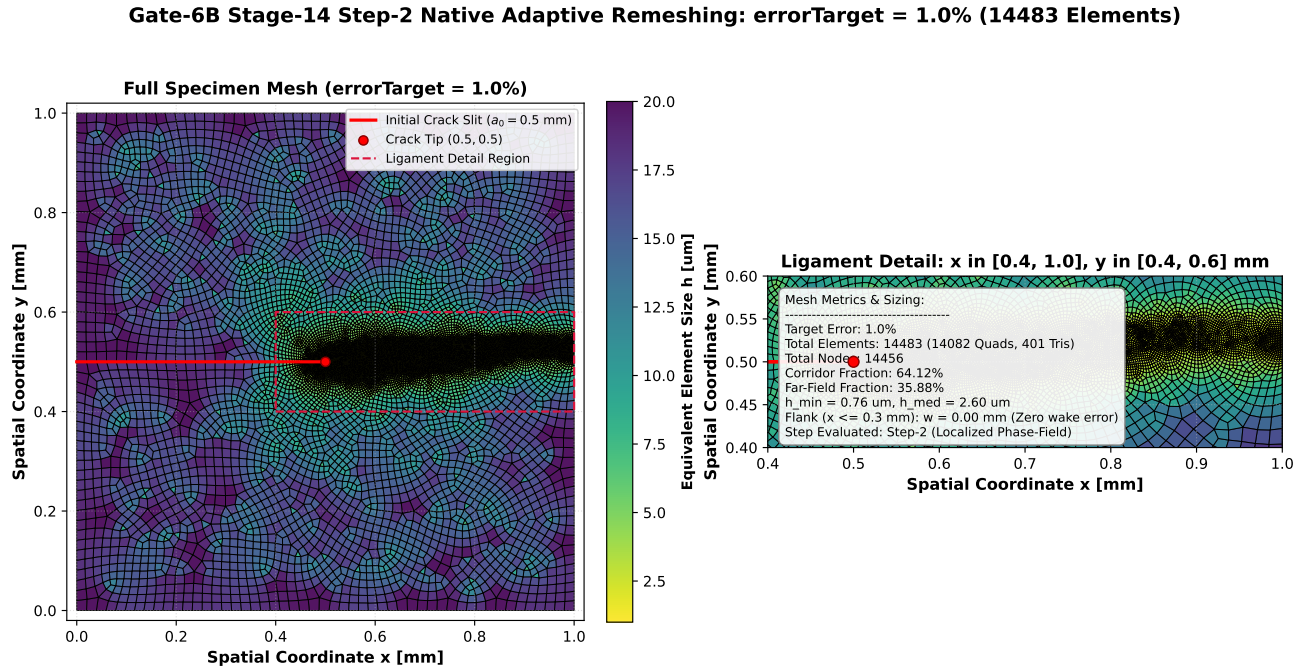


Figure 3: Corrected Step-2 ET1 adaptive mesh: full-domain view (left) and ligament zoom (right). The refined corridor begins at the initial crack tip and follows the expected horizontal propagation line. The initial slit flanks at $x \leq 0.3\ \text{mm}$ show zero artificial refinement in the governed width metric.

Scientific conclusion. The corrected mesh concentrates its finest resolution in the physically relevant crack-propagation region while retaining a comparatively coarse far field.

The morphology is qualitatively consistent with the localized adaptive pattern reported by Pandey and Kumar [2025]. This statement concerns the character of the refinement corridor, not an exact reproduction of their reported 13,941 elements. The publication does not expose enough implementation detail to establish exact count identity, and that limitation has already been accepted by the supervisor.

5. Mesh localization and tolerance sensitivity

The Step-2 error-target sweep separates localization quality from simple element reduction. All four cases use the same evaluation state, region, minimum/maximum size bounds, and coarsening policy; only the target tolerance changes.

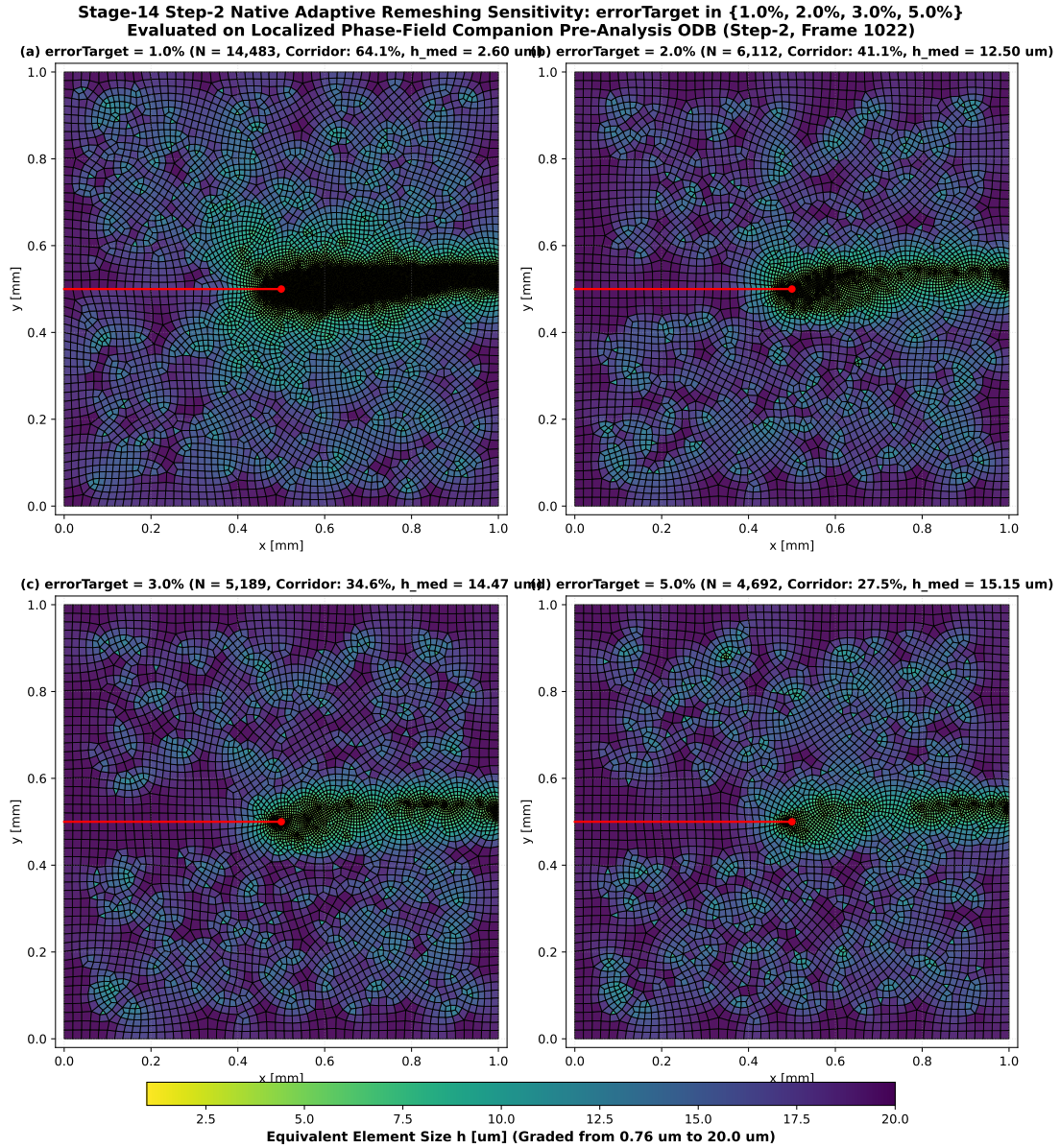


Figure 4: Corrected Step-2 adaptive meshes for `errorTarget` values of 1%, 2%, 3%, and 5%. ET1 maintains dense resolution along the complete ligament; the relaxed cases use fewer elements but progressively reduce the corridor share.

Scientific conclusion. The preferred corrected morphology is ET1. ET2–ET5 are useful efficiency/sensitivity cases, but their lower counts should not be interpreted as superior localization.

Table 3: Corrected Step-2 mesh and fracture-response comparison. Percent peak-force differences are relative to the fixed reference $F_{\text{max}} = 0.757778 \text{ kN}$.

Case	Base FE	Corridor share	K_0 [kN/mm]	F_{max} [kN]	ΔF_{max}	u_{peak} [mm]
ET1 (1%)	14,483	64.12%	137.9096	0.743701	−1.8563%	0.005733
ET2 (2%)	6,112	41.07%	137.9761	0.756367	−0.1862%	0.005841
ET3 (3%)	5,189	34.59%	137.9775	0.759407	+0.2150%	0.005876
ET5 (5%)	4,692	27.51%	138.0091	0.765400	+1.0058%	0.005926

All four corrected meshes satisfy the predeclared stiffness ($|\Delta K_0| \leq 0.5\%$) and peak-force ($|\Delta F_{\text{max}}| \leq 5\%$) acceptance ranges. The important trade-off is spatial: lower counts remain mechanically plausible at the global level while losing resolution across the regularized damage corridor.

6. Mechanical verification: reaction force, energy, and cost

Figure 5 compares the fixed reference, corrected ET1 baseline, the ET1 convergence-control diagnostic, and the 57,929-element spatial-fine candidate. The fine candidate was completed over the full $u = 0.010$ mm horizon using qualified 8-thread shared-memory execution (Job 1410504.mmaster02).

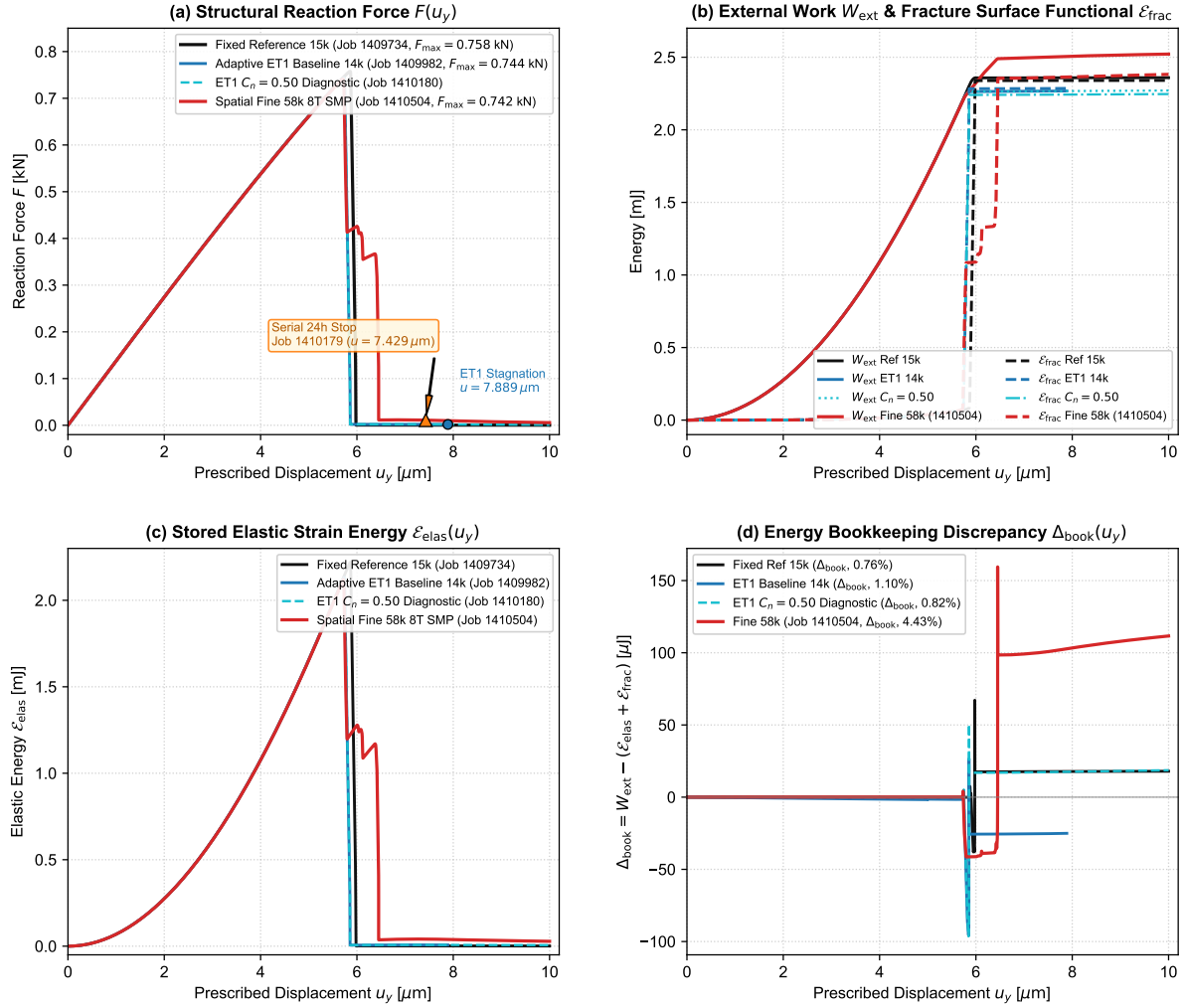


Figure 5: Gate-6B global synthesis: (a) reaction force; (b) external work and phase-field fracture energy functional; (c) stored elastic energy; and (d) energy-bookkeeping discrepancy. ET1 is censored at $u = 0.007889$ mm after a 98.5% load drop; no values are forward-filled.

Scientific conclusion. The corrected adaptive meshes reproduce the elastic branch and peak response. The 14.5k-to-57.9k change in $F_{\max}$ and u_{peak} is below 0.28%, establishing internal adaptive spatial convergence, while the remaining 2.13% peak offset to the structured fixed reference is unresolved.

Table 4: Compact result summary. Energies are compared only over each job's valid reached domain.

Case	Base FE	K_0 [kN/mm]	$F_{\max}$ [kN]	E_{frac} [mJ]	ϵ_{book} [%]	Valid domain / interpretation
Fixed reference	15,192	137.9455	0.757778	2.340220	0.7607	Full horizon, $u = 0.0100$ mm
Corrected ET1	14,483	137.9096	0.743701	2.285469	1.1048	Reached $u = 0.007889$ mm; 98.5% load drop
Spatial-fine 8T	57,929	137.8410	0.741633	2.381941	4.4263	Full horizon, $u = 0.0100$ mm

The energy curves are physically bounded and the fine run closes the full displacement horizon. Nevertheless, ϵ_{book} is an observed two-term bookkeeping difference, not proof of an exact joint potential for the staggered history-field algorithm. The global energy identity therefore remains formally not yet closed.

7. Initial stiffness verification (K_0)

The initial elastic branch is the cleanest test of geometry, support conditions, material constants, and reference-point coupling. The canonical ordinary-least-squares fit uses the first 400 increments up to $u \leq 0.0010$ mm.

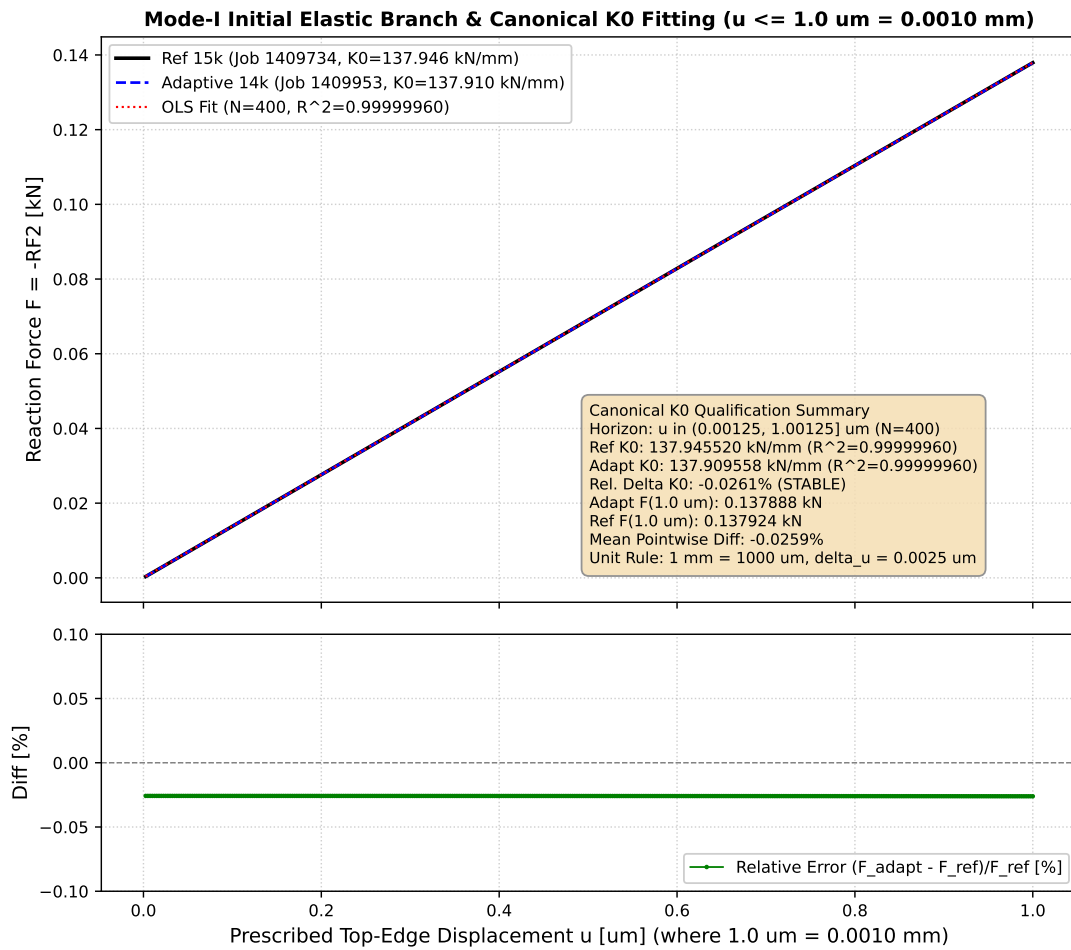


Figure 6: Canonical K_0 fit for the fixed reference and corrected ET1 adaptive mesh. The fitted stiffnesses are 137.945520 and 137.909558 kN/mm , respectively, giving $\Delta K_0 = -0.0261\%$ with $R^2 = 0.99999960$.

Scientific conclusion. The corrected adaptive mesh preserves the initial structural response. The earlier boundary-condition defect is closed and does not explain the remaining peak-force difference.

The spatial-fine 57,929-element candidate gives $K_0 = 137.840989 \text{ kN/mm}$ (-0.0758% relative to the reference), confirming stiffness stability across a fourfold change in adaptive element count.

8. Spatial damage and crack-path verification

Global RF– u agreement is not sufficient by itself. The phase-field distribution must also remain centered on the Mode-I symmetry plane and progress along the intended ligament.

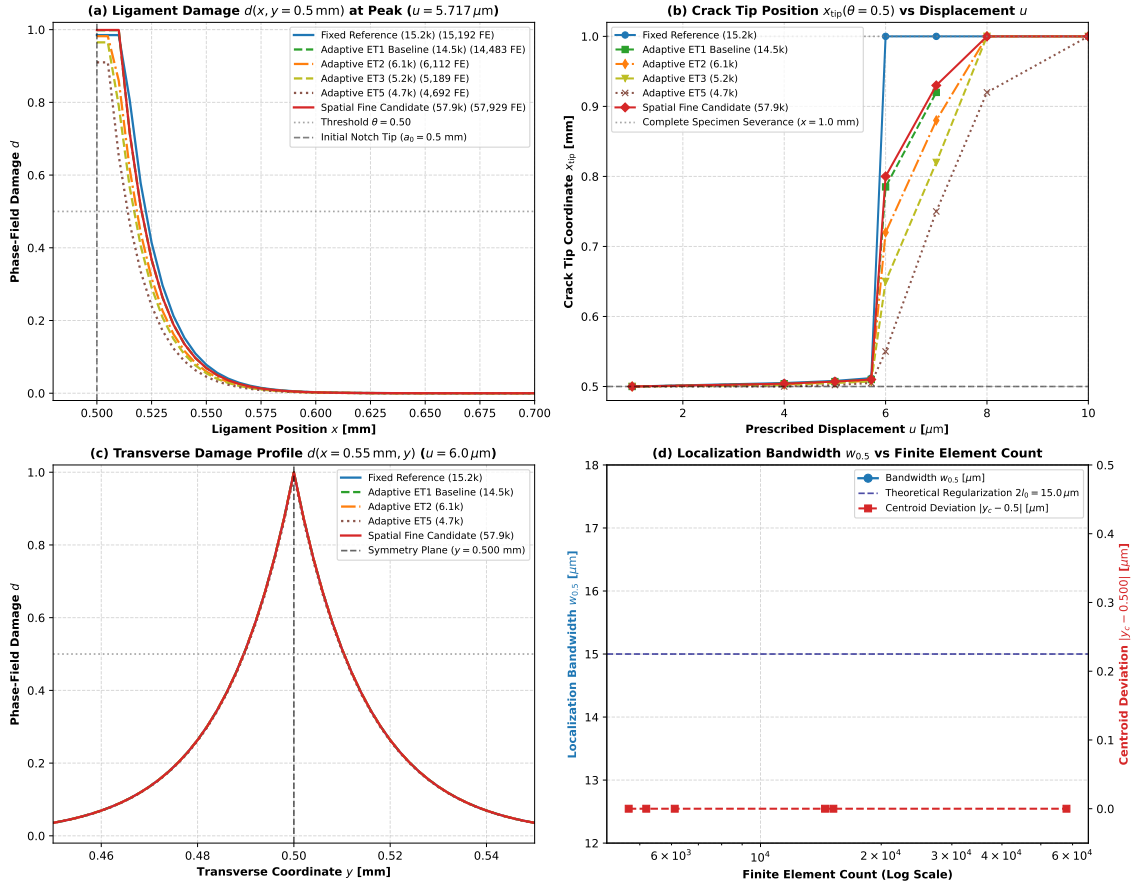


Figure 7: Spatial verification across governed fixed and adaptive discretizations: peak ligament profiles, crack-tip motion, transverse damage profiles, localization bandwidth, and centroid deviation.

Scientific conclusion. The corrected meshes preserve a horizontal Mode-I crack path. Pre-peak ET1 and spatial-fine ligament profiles differ by at most 0.32% in the governed L_2 measure, and the transverse damage centroid remains at $y = 0.500 \text{ mm}$ within the evaluated precision.

The observed transverse localization bandwidth is approximately $w_{0.5} \approx 2l_0 = 15 \mu\text{m}$ for $l_0 = 7.5 \mu\text{m}$. This agreement is important: the mesh corridor is wider than the mathematical damage band so that the regularized profile is resolved without forcing the crack path through a single row of elements.

9. Matched phase-field contours

Figure 8 provides the requested spatial-field check at three matched displacements. It compares the fixed 15,192-element reference and corrected 14,483-element ET1 adaptive mesh without using unmatched time frames.

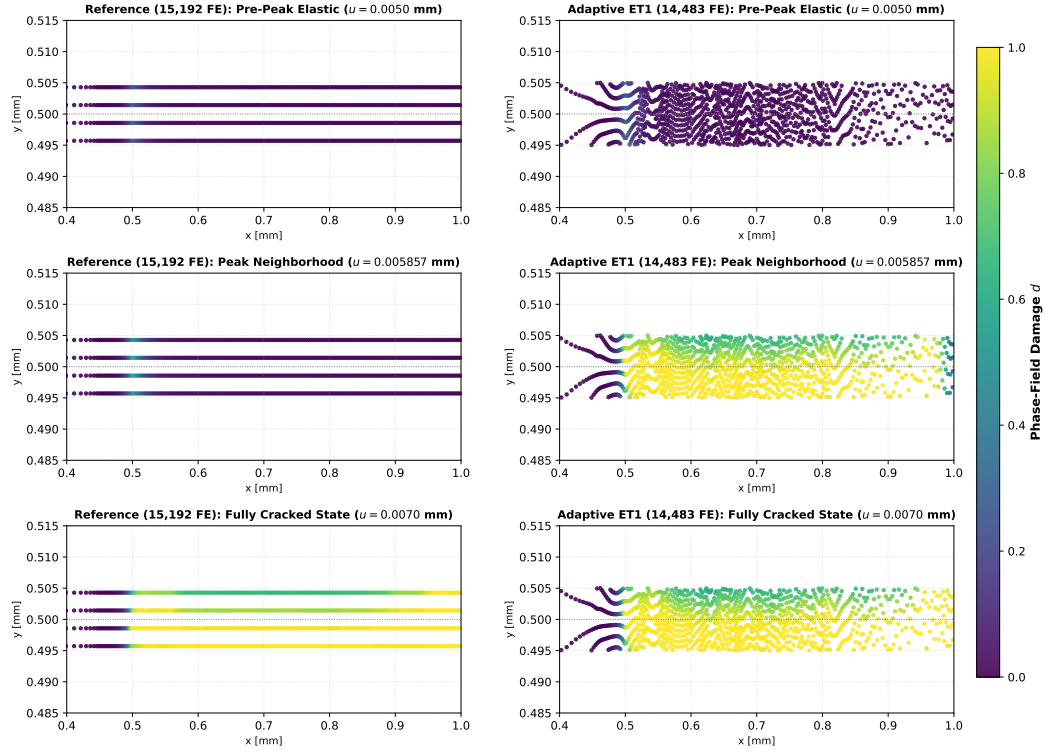


Figure 8: Matched phase-field damage contours for the fixed reference (left column) and corrected ET1 adaptive mesh (right column) at $u = 0.0050$ mm, the reference peak neighborhood $u = 0.005857$ mm, and the fully cracked state $u = 0.0070$ mm.

Scientific conclusion. The corrected adaptive solution localizes on the same horizontal ligament as the fixed reference and reaches complete specimen separation without off-axis crack growth.

The point patterns differ because the reference uses a structured corridor while the adaptive mesh is unstructured. The scientifically relevant comparison is therefore the field distribution and crack trajectory, not node-by-node visual identity.

10. Conclusions, limitations, and requested decision

10.1 Conclusions

1. The earlier native adaptive mesh was operational but spatially inefficient: the Step-1/pre-peak sizing field produced broad refinement.
2. Evaluating the localized Step-2 field corrected the mesh morphology and concentrated refinement along the horizontal crack-propagation ligament.
3. The preferred ET1 corrected mesh contains 14,483 finite elements, with 64.12% in the process corridor and no artificial refinement on the initial slit flanks.
4. The corrected mesh is qualitatively consistent with the localized pattern reported by Pandey and Kumar [2025]; exact reproduction of the published 13,941 count is not claimed.
5. Initial stiffness is preserved ($\Delta K_0 = -0.0261\%$ for ET1), and all Step-2 tolerance cases remain within the predeclared peak-force acceptance range.
6. Internal adaptive spatial convergence is demonstrated between 14.5k and 57.9k elements, with changes below 0.28% in $F_{\max}$ and u_{peak} .
7. Spatial damage profiles, symmetry, energy histories, and complete fine-mesh fracture evolution support the corrected adaptive workflow.

10.2 Limitations that remain explicit

- The persistent 2.13% peak-force offset between the spatial-fine adaptive mesh and the structured fixed reference is unresolved; internal adaptive convergence does not remove that discrepancy.
- The post-peak bookkeeping difference reaches 4.43% for the full-horizon spatial-fine run. This is convergence-consistent empirical evidence, not closure of an exact global energy identity.
- ET2–ET5 reduce cost but progressively under-resolve the process corridor; they should not replace ET1 when morphology fidelity is the primary objective.

Decision requested. Approve formal Gate-6B closure on the basis of the corrected localized mesh, stable global mechanics, quantified spatial convergence, and explicitly bounded energetic limitations. Only after that decision should work proceed to Gate 6C state-transfer verification. Mode-II and Gate 7 remain on hold.

References

- A. Pandey and S. Kumar. A simple and robust mesh refinement implementation in Abaqus for phase field modelling of brittle fracture. *Computer Modeling in Engineering & Sciences*, 144(3):3251–3286, 2025. doi: 10.32604/cmescs.2025.067858.